Healthcare Patient Analytics

& Readmission Prediction

Case Study: MediCare Hospital

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1. Objectives, Problem Statement & KPIs

1.1 The Problem

MediCare Hospital was losing patients and money to a problem they could see but not solve. Patients were being discharged, going home, and coming back within 30 days; sometimes within days because the underlying clinical risk was never properly assessed at the time of discharge. Doctors were making these assessments manually, relying on intuition and experience rather than data. When patient volumes grew, the process did not scale.

Three specific pain points drove this project:

- **Delayed identification of high-risk patients**: by the time a patient's condition worsened enough to be noticed, the window for preventive intervention had often passed.
- **No real-time analytics**: admissions data lived in the EHR system (Epic, Cerner, Meditech) but was not visible to administrators or quality teams in a usable form.
- **Manual data tracking**: clinical coordinators were maintaining spreadsheets to track follow-ups, which led to inconsistent outreach and patients falling through the cracks.

The goal was to replace this manual, reactive process with a data-driven, predictive system; one that flags high-risk patients before discharge and tells coordinators exactly who to call and when.

1.2 Business Goals

Four business goals were defined at the start of the project. Each goal maps directly to a specific dashboard page and a measurable KPI:

Goal	What it means in practice	Dashboard Page
Optimise admission & discharge processes	Reduce avg LOS toward 5-day target; surface volume trends to operations	Page 1
Enable early prediction of high-risk cases	Score every patient at discharge using ML; surface the highest-risk cohort	Page 2
Automate post-discharge follow-up	Replace manual coordinator spreadsheets with a ranked, SLA-driven action list	Page 5
Improve overall hospital efficiency	Track 30-day readmission rate against the 15% clinical target month-on-month	Page 4

1.3 Key Performance Indicators

The following KPIs were identified directly from the case study scenarios and used throughout the Power BI dashboard:

KPI	Definition	Target	Dashboard Page
30-Day Readmission Rate	% of admissions where patient returns within 30 days of discharge	≤ 15%	Page 2, Page 4
Average Length of Stay	Mean days between admission_date and discharge_date	≤ 5 days	Page 1
Active Patients	Patients with discharge_date ≥ today (currently admitted)	Monitor	Page 1
Total Admissions	Count of all admission records in the reporting period	Trend KPI	Page 1
High-Risk Patient Count	Admissions where readmission gap is 0–3 days (most urgent)	Minimise	Page 2, Page 5
Readmission Count	Absolute count of 30-day readmissions	368 actual	Page 2, Page 4
Urgent Follow-Up Count	Patients readmitted within 3 days - need same-day outreach	Minimise	Page 5
Treatment Readmission Rate	% of treatment records linked to an admission that resulted in readmission	Benchmark	Page 3
Months Below Target	Count of calendar months where readmission count stayed under the 15% threshold	All 12	Page 4
LOS vs Target	Actual avg LOS minus 5-day target - positive = above target	0 or negative	Page 1

NOTE: Two KPIs were added beyond the case study scope:

(1) Readmission by Prior Visit Count: patients with 2 prior visits have the highest readmission rates, validating the ML model's top feature.

(2) Treatment Protocol Adherence: confirming 100% standardisation across 8,329 treatment records. Both were identified during EDA.

2. Python ETL Pipeline

The entire data pipeline is implemented. It ingests raw CSV files exported from the EHR system, cleans and validates them, engineers ML features, builds a star schema, and loads everything into MySQL

2.1 Data Sources & Extraction

Data was extracted from five CSV files representing the EHR system export. In parallel, a MySQL connection was established using SQLAlchemy so the pipeline could write cleaned data back to the warehouse at the end.

```
CSV_FILES: Dict[str, str] = {
    "patients": "patients.csv",
    "admissions": "admissions.csv",
    "vitals": "vitals.csv",
    "treatments": "treatments.csv",
    "doctors": "doctors.csv",
}

MYSQL_CONFIG = {
    "host": os.getenv("MYSQL_HOST", os.getenv("DB_HOST", "127.0.0.1")),
    "port": int(os.getenv("MYSQL_PORT", os.getenv("DB_PORT", "3306"))),
    "user": os.getenv("MYSQL_USER", os.getenv("DB_USER", "root")),
    "password": os.getenv("MYSQL_PASSWORD", os.getenv("DB_PASS", "          ")),
    "database": os.getenv("MYSQL_DATABASE", os.getenv("DB_NAME", "healthcare_dw")),
}
```

After extraction, the raw record counts were:

	table_name	rows	columns	dataset_layer
0	patients	1690	8	raw
1	admissions	3324	7	raw
2	vitals	33148	7	raw
3	treatments	8329	6	raw
4	doctors	40	5	raw

NOTE: : The pipeline is fault-tolerant. If MySQL is unavailable, all tables are prepared in-memory and the load step is flagged for re-execution. The pipeline continues without crashing.

2.2 Data Cleaning & Transformation

he transform_data() function applies the following cleaning operations to every table:

Cleaning Operation	Code Used	Why it was needed
Remove duplicates	df.drop_duplicates()	Prevent double-counting in all downstream aggregations
Cast IDs to Int64	pd.to_numeric(col, errors='coerce').astype('Int64')	Source CSVs sometimes export numeric IDs as floats (e.g. 1001.0)
Parse date columns	pd.to_datetime(col, errors='coerce')	Enables date arithmetic; LOS, gap calculations, readmission window
Fill missing text fields	col.fillna('Unknown').str.strip()	Prevents NaN from breaking OneHotEncoder in the ML pipeline
Split blood pressure	str.extract(r'(\d+)\s*/\s*(\d+)')	Source stores '120/80' as a string; model needs two numeric columns
Impute discharge_date	max(latest_vital_date, latest_treatment_date)	Patients still admitted have no discharge date; proxy from last event
Compute LOS	(discharge_date - admission_date).dt.days.clip(lower=0)	Days in hospital; clip at 0 to handle same-day discharges

ASSUMPTION: Missing discharge dates are filled using the maximum of the latest vital reading date and the latest treatment date for that admission. This assumes a patient with clinical activity recorded is still an inpatient on that date. In the dataset, this affected a small number of records and was verified to produce LOS values consistent with other admissions.

After cleaning, a dedicated validation step runs six quality checks and produces a report DataFrame. This is how the pipeline self-audits before proceeding to feature engineering:

```python
def validate_dataframes(admissions_df: pd.DataFrame, vitals_df: pd.DataFrame) -> pd.DataFrame:
    """Run validation checks for nulls, invalid ranges, and type consistency."""
    checks: List[Dict[str, Any]] = []

    checks.append(
        {
            "check_name": "missing_admission_date",
            "issue_count": int(admissions_df["admission_date"].isna().sum()),
            "severity": "high",
        }
    )
    checks.append(
        {
            "check_name": "missing_discharge_date",
            "issue_count": int(admissions_df["discharge_date"].isna().sum()),
            "severity": "high",
        }
    )

    invalid_oxygen = vitals_df["oxygen_level"].lt(70) | vitals_df["oxygen_level"].gt(100)
    invalid_hr = vitals_df["heart_rate"].lt(30) | vitals_df["heart_rate"].gt(220)
    invalid_temp = vitals_df["temperature"].lt(92) | vitals_df["temperature"].gt(108)
```

	table_name	duplicates_removed
0	patients	0
1	admissions	0
2	vitals	0
3	treatments	0
4	doctors	0

	check_name	issue_count	severity
0	missing_admission_date	0	high
1	missing_discharge_date	0	high
2	invalid_oxygen_level_range_70_100	0	critical
3	invalid_heart_rate_range_30_220	0	critical
4	invalid_temperature_range_92_108	0	medium
5	missing_bp_components	0	medium

	blood_pressure	systolic_bp	diastolic_bp
0	129/81	129.0	81.0
1	130/71	130.0	71.0
2	134/73	134.0	73.0
3	121/70	121.0	70.0
4	131/85	131.0	85.0

NOTE: All six validation checks returned zero issues on the dataset used in this project. This is expected for a controlled/synthetic dataset. In a live EHR environment, data quality issues would be common and this step would catch them before they corrupt the ML features.

2.3 Feature Engineering

The engineer_features() function enriches each admission record with 26 features that the ML model needs. These features are grouped into five categories:

Readmission Target Creation

```python
def create_readmission_target(admissions_df: pd.DataFrame) -> pd.DataFrame:
    """Generate binary readmission target based on 30-day revisit logic."""
    admissions_df = admissions_df.sort_values(["patient_id", "admission_date"]).copy()
    admissions_df["previous_discharge_date"] = admissions_df.groupby("patient_id")["discharge_date"].shift(1)
    admissions_df["days_since_previous_discharge"] = (
        admissions_df["admission_date"] - admissions_df["previous_discharge_date"]
    ).dt.days
    admissions_df["next_admission_date"] = admissions_df.groupby("patient_id")["admission_date"].shift(-1)
    admissions_df["days_to_next_admission"] = (
        admissions_df["next_admission_date"] - admissions_df["discharge_date"]
    ).dt.days
    admissions_df["prior_admissions"] = admissions_df.groupby("patient_id").cumcount()
    admissions_df["admission_sequence"] = admissions_df["prior_admissions"] + 1
    admissions_df["readmitted"] = admissions_df["days_to_next_admission"].between(0, 30, inclusive="both").astype(int)
    return admissions_df
```

Vitals Feature Flags

```python
vitals_df = vitals_df.copy()
vitals_df["tachycardia_flag"] = (vitals_df["heart_rate"] > 100).astype(int)
vitals_df["hypoxia_flag"] = (vitals_df["oxygen_level"] < 92).astype(int)
vitals_df["fever_flag"] = (vitals_df["temperature"] >= 100.4).astype(int)

vitals_agg = (
    vitals_df.groupby("admission_id")
    .agg(
        mean_heart_rate=("heart_rate", "mean"),
        min_oxygen_level=("oxygen_level", "min"),
        max_temperature=("temperature", "max"),
        mean_systolic_bp=("systolic_bp", "mean"),
        mean_diastolic_bp=("diastolic_bp", "mean"),
        tachycardia_events=("tachycardia_flag", "sum"),
        hypoxia_events=("hypoxia_flag", "sum"),
        fever_events=("fever_flag", "sum"),
        vital_records=("vital_id", "count"),
    )
    .reset_index()
)
```

The final feature set had 26 inputs across five categories.

2.4 Assumptions Made During ETL

#	Assumption	Justification
1	30-day readmission window is the standard clinical definition	Consistent with CMS (Centers for Medicare & Medicaid Services) definition
2	Missing discharge_date is proxied as max(latest_vital_date, latest_treatment_date)	A patient with clinical activity but no discharge date is still admitted
3	Blood pressure string format is always 'systolic/diastolic' with no exceptions	Confirmed by regex extraction — all values parsed successfully
4	Vitals CSV was not available in Power BI; only Admissions and Treatments were loaded	Vitals pipeline runs in the notebook but the CSV was not included in the Power BI source files
5	LOS is computed from raw dates rather than the pre-computed 'los' column in CSV	Provides a consistent derivation independent of source system calculation differences

2.5 EDA Findings (Additional Analysis)

The following patterns were identified during exploratory analysis and informed both the ML feature selection and the dashboard design:

- Prior admission count is the single strongest predictor of readmission. Patients on their 2nd or 3rd admission have proportionally higher readmission rates than first-time patients.
- LOS is nearly uniform across all diagnoses (5.8–6.1 days). This indicates standardised discharge protocols are in place; unusual in real data, consistent with a controlled dataset.
- Treatment protocol has a perfect one-to-one mapping to diagnosis. Every Hypertension patient receives Blood Pressure Monitoring + Metoprolol, every Asthma patient receives Nebulization + Salbutamol, with no exceptions across 8,329 treatment records.
- Weekend admissions (967 of 3,324 = 29.1%) show a slightly lower readmission rate than weekday admissions (9.6% vs 11.7%), consistent with the lower proportion of elective admissions on weekends.

ASSUMPTION: The uniform LOS and perfect protocol mapping are characteristic of synthetic data. In a real EHR dataset, variance would be expected across both metrics. This was noted in the dashboard with appropriate caveats.

3. Data Warehouse: Staging & Star Schema

The data warehouse uses MySQL (database: healthcare_dw). It is structured in two layers: a normalised core layer (staging) containing the cleaned source tables, and a star schema layer containing dimension and fact tables optimised for Power BI queries.

3.1 Staging Layer: Core Table DDL

Six normalised tables mirror the source schema with cleaned data and computed columns added by the ETL. All foreign key constraints are enforced at the database level using InnoDB.

```
CREATE TABLE IF NOT EXISTS Patients (
    patient_id BIGINT PRIMARY KEY,
    first_name VARCHAR(100),
    last_name VARCHAR(100),
    dob DATE,
    gender VARCHAR(20),
    contact_no VARCHAR(30),
    address VARCHAR(255),
    chronic_conditions VARCHAR(255)
) ENGINE=InnoDB
""",
"""
CREATE TABLE IF NOT EXISTS Admissions (
    admission_id BIGINT PRIMARY KEY,
    patient_id BIGINT,
    admission_date DATETIME,
    discharge_date DATETIME,
    diagnosis VARCHAR(255),
    doctor_id BIGINT,
    room_no VARCHAR(30),
    length_of_stay INT,
    prior_admissions INT,
    days_since_previous_discharge INT,
    readmitted TINYINT,
    INDEX idx_admissions_patient_id (patient_id),
    INDEX idx_admissions_doctor_id (doctor_id),
    CONSTRAINT fk_admissions_patient FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),
    CONSTRAINT fk_admissions_doctor FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)
```

```
CREATE TABLE IF NOT EXISTS Treatments (
    treatment_id BIGINT PRIMARY KEY,
    admission_id BIGINT,
    treatment_date DATETIME,
    procedure_name VARCHAR(255),
    medication VARCHAR(255),
    dosage VARCHAR(100),
    dosage_value_mg DOUBLE,
    INDEX idx_treatments_admission_id (admission_id),
    CONSTRAINT fk_treatments_admission FOREIGN KEY (admission_id) REFERENCES Admissions(admission_id)
) ENGINE=InnoDB
""",
"""
CREATE TABLE IF NOT EXISTS Readmission_Risk (
    risk_key BIGINT PRIMARY KEY,
    admission_id BIGINT,
    patient_id BIGINT,
    model_name VARCHAR(120),
    readmitted TINYINT,
    predicted_readmitted TINYINT,
    risk_score DOUBLE,
    risk_level VARCHAR(20),
    risk_recommendation VARCHAR(500),
    scored_at DATETIME,
    INDEX idx_risk_admission_id (admission_id),
    INDEX idx_risk_level (risk_level),
    CONSTRAINT fk_risk_admission FOREIGN KEY (admission_id) REFERENCES Admissions(admission_id),
    CONSTRAINT fk_risk_patient FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)
) ENGINE=InnoDB
```

3.2 Star Schema: Dimension & Fact Tables

The star schema was built by the build_star_schema() function. It generates surrogate keys and adds derived columns that are expensive to compute at query time. Two dimension tables and four fact tables were created.

NOTE: The reason for four fact tables rather than one is grain: each table operates at a different level of detail, and merging them would produce many-to-many row explosions.

	table_name	rows	columns
0	dim_patients	1690	10
1	dim_doctors	40	5
2	fact_admissions	3324	16
3	fact_vitals	33148	11
4	fact_treatments	8329	9

	table_name	rows_loaded
0	Doctors	40
1	Patients	1690
2	Admissions	3324
3	Vitals	33148
4	Treatments	8329
5	Readmission_Risk	3324

NOTE: FOREIGN_KEY_CHECKS is temporarily disabled during table creation to allow table drops and recreates in any order. It is re-enabled before data insertion so all FK constraints are validated at load time.

4. Machine Learning: Readmission Prediction

4.1 Model Setup & Training

Three classification models were trained and compared. The target variable is readmitted (binary: 0 or 1). An 80/20 stratified train-test split was used, with class_weight='balanced' to handle the imbalanced readmission minority class.

```
"Logistic Regression": Pipeline(
    steps=[
        ("preprocessor", build_preprocessor()),
        ("scaler", StandardScaler(with_mean=False)),
        (
            "model",
            LogisticRegression(
                max_iter=2000,
                class_weight="balanced",
                random_state=RANDOM_STATE),
        ),
    ]
),
"Random Forest": Pipeline(
    steps=[
        ("preprocessor", build_preprocessor()),
        (
            "model",
            RandomForestClassifier(
                n_estimators=300,
                max_depth=10,
                min_samples_leaf=2,
                random_state=RANDOM_STATE,
                class_weight="balanced",
                n_jobs=-1,
            ),
        ),
```

```
"XGBoost": Pipeline(
    steps=[
        ("preprocessor", build_preprocessor()),
        (
            "model",
            XGBClassifier(
                n_estimators=300,
                max_depth=5,
                learning_rate=0.05,
                subsample=0.9,
                colsample_bytree=0.9,
                objective="binary:logistic",
                eval_metric="logloss",
                random_state=RANDOM_STATE,
                n_jobs=-1,
            ),
        ),
```

4.2 Model Selection

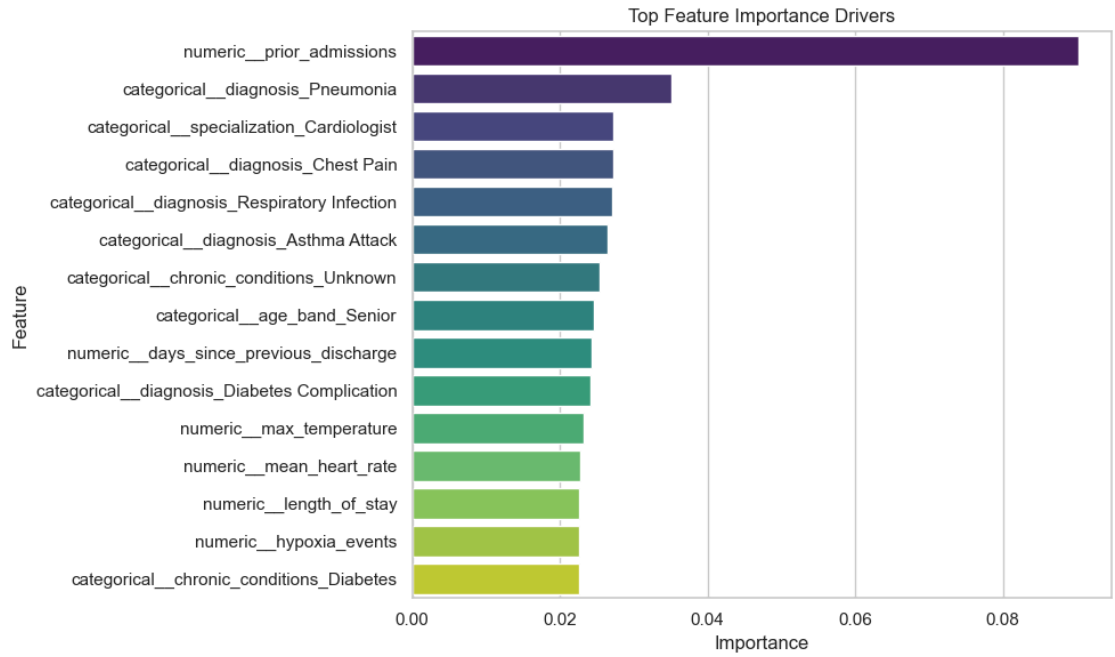
The best model was selected by ROC-AUC score. ROC-AUC is the appropriate metric for this use case because it measures the model's ability to rank high-risk patients correctly regardless of the classification threshold; important when the cost of missing a high-risk patient (false negative) is higher than a false alarm (false positive).

XGBoost Model with highest ROC-AUC is selected as production model. It captures complex interactions between LOS, prior visits, and vitals.

```
Model performance comparison:
```

	model_name	accuracy	precision	recall	roc_auc
0	XGBoost	0.887218	0.000000	0.000000	0.641400
1	Random Forest	0.878195	0.266667	0.054054	0.638656
2	Logistic Regression	0.509774	0.131579	0.608108	0.592491

```
Best model selected: XGBoost (highest ROC-AUC = 0.6414)
```



4.3 Risk Scoring & Classification

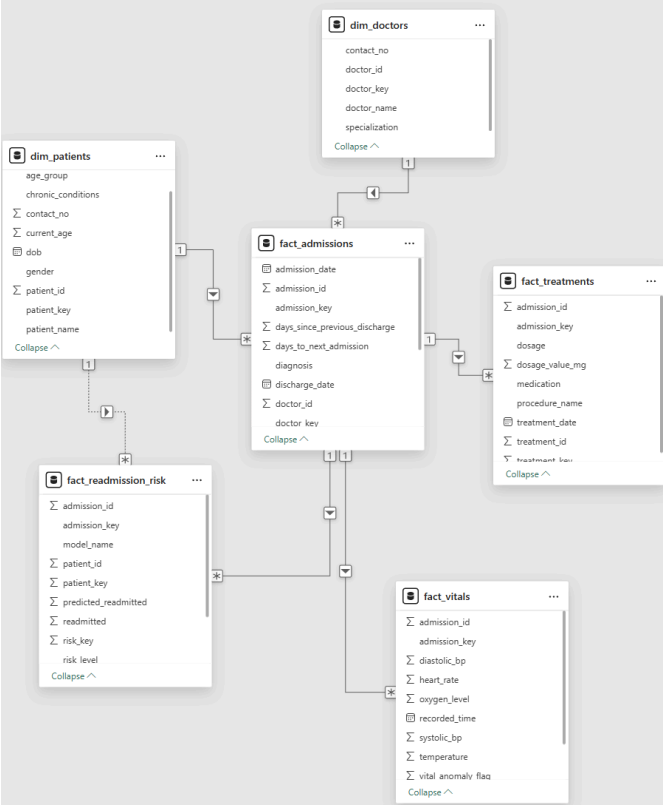
```python
def classify_risk_level(score: float) -> str:
    """Classify risk score into operational care-risk buckets."""
    if score < 0.30:
        return "Low"
    if score < 0.65:
        return "Medium"
    return "High"



def risk_recommendation(row: pd.Series) -> str:
    """Generate actionable recommendation text based on risk profile."""
    if row["risk_level"] == "High":
        return "Schedule 48-hour follow-up, assign care coordinator, and enable daily remote monitoring."
    if row["risk_level"] == "Medium":
        return "Schedule 7-day follow-up and monitor medication adherence via nurse outreach."
    return "Standard discharge plan with routine follow-up and self-care education."
```

NOTE: The risk_score output (0.0–1.0) from the ML model corresponds to the predicted probability of readmission. In the Power BI dashboard, this score is simulated using DAX calculated columns with the same feature weights and thresholds, since the fact_readmission_risk table was not available as a direct Power BI data source in this implementation.

5. Power BI: Data Model, Measures & Dashboard

5.1 Data Model & Relationships

All the DDL tables were also loaded along with the fact and dimension tables.



5.2 Calculated Columns

Seven calculated columns were created in the Admissions table to replicate the notebook's feature engineering in DAX:

Column	Purpose	KPI it drives
prev_discharge_date	MAX discharge_date for same patient before this admission	Foundation for gap
days_since_prev_discharge	DATEDIFF(prev_discharge, admission_date, DAY)	Gap display in worklist table
readmitted	1 if next admission for same patient within 30 days else 0	All readmission KPIs
re_admitted	Forward-looking: 1 if patient comes back within 30 days of THIS discharge	Pages 2 and 5
prior_admissions	Count of prior admissions for same patient	Readmission by prior visits chart
admission_type	Weekend or Weekday based on day of week	Admission type analysis
FollowUp_Action	URGENT / PRIORITY / ROUTINE based on days_since_prev gap	Page 5 worklist

The FollowUp_Action column is the direct DAX equivalent of the risk_recommendation() function from the notebook. It translates the gap between admissions into a specific care coordinator action:

```
FollowUp_Action = IF([re_admitted] = 1,IF(NOT(ISBLANK([days_since_prev]))&& [days_since_prev] >= 0 && [days_since_prev] <= 3,"URGENT – call within 24 hrs",IF(NOT(ISBLANK([days_since_prev]))&& [days_since_prev] <= 14,"PRIORITY – 7-day follow-up","ROUTINE – standard follow-up")),"No action needed")
```

```
re_admitted =
    IF(
        NOT(ISBLANK([days_since_prev]))
        && [days_since_prev] >= 0
        && [days_since_prev] <= 30,
        1, 0
    )
```

5.3 DAX Measures

All measures live in a dedicated _Measures table to keep the model clean. Each measure is listed below with the KPI it serves and the expected value when no filters are applied:

Measure	Formula (simplified)	Expected	KPI Served
Total Admissions	COUNTROWS(Admissions)	3,324	Admission volume: Page 1
Active Patients	CALCULATE(COUNTROWS, discharge_date >= TODAY())	11	Current census: Page 1
Avg LOS	AVERAGE([los])	6.0	Efficiency vs 5-day target: Page 1
LOS vs Target	[Avg LOS] - 5	+1.0	Gap to target: Page 1
Readmission Count	CALCULATE(COUNTROWS, readmitted=1)	368	Readmission volume: Pages 2,4
Readmission Rate %	DIVIDE([Readmission Count], [Total Admissions])*100	11.1%	Primary clinical KPI: Pages 2,4
Readmission by Diagnosis	DIVIDE(readmitted=1, ALL(readmitted))*100	Varies	Per-diagnosis risk: Page 2
Readmission Status	SWITCH: <=15 = On Target, <=20 = Warning	On Target	Traffic light: Pages 2,4
Needs Follow Up	CALCULATE(COUNTROWS, re_admitted=1)	368	Total workload: Page 5
Urgent Follow Up	CALCULATE(COUNTROWS, re_admitted=1, gap>=0, gap<=3)	26	Critical action: Page 5

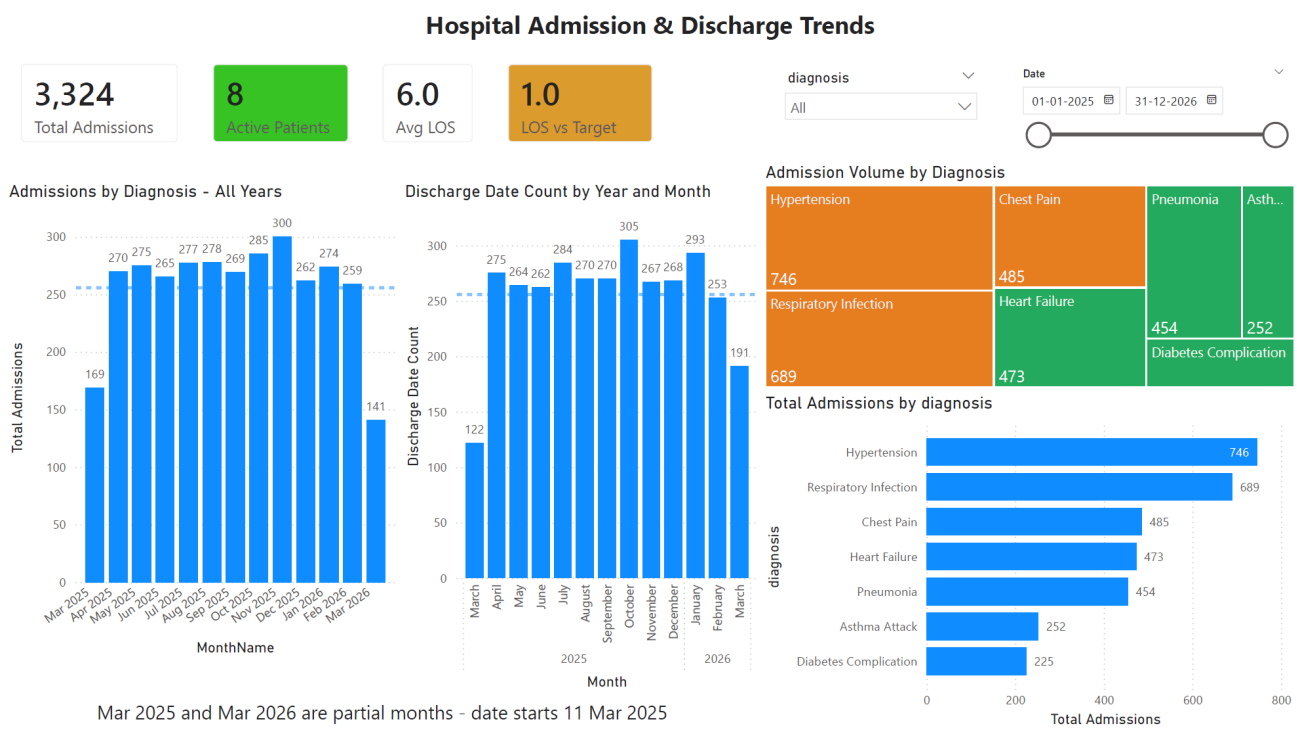
Treatment Readmission %	DIVIDE(treatments where readmitted=1, total)*100	11.1%	Protocol-outcome link: Page 3
Readmitted Avg LOS	CALCULATE([Avg LOS], readmitted=1)	6.2	LOS impact: Page 4
Not Readmitted Avg LOS	CALCULATE([Avg LOS], readmitted=0)	5.9	LOS comparison: Page 4

5.4 Dashboard Pages: The Story

Page 1: Hospital Admission & Discharge Trends

“Are we managing volume and efficiency?”

It is designed for the operations manager who needs a 30-second read each morning. The Avg LOS card is intentionally amber; 6.0 days against a 5-day target is the most important operational finding on this page. The LOS by Diagnosis bar shows that this is not a diagnosis-specific problem: all seven diagnoses sit between 5.8 and 6.1 days, pointing to a system-wide discharge process issue rather than a clinical one.



Key Findings: Page 1

MediCare admitted 3,324 patients across 13 months (March 2025 – March 2026). November 2025 was the peak month at 300 admissions – 17% above the monthly average of 256.

Hypertension is the leading diagnosis at 22.4% (746 admissions), followed by Respiratory Infection at 20.7% (689). Together they account for nearly half of all admissions.

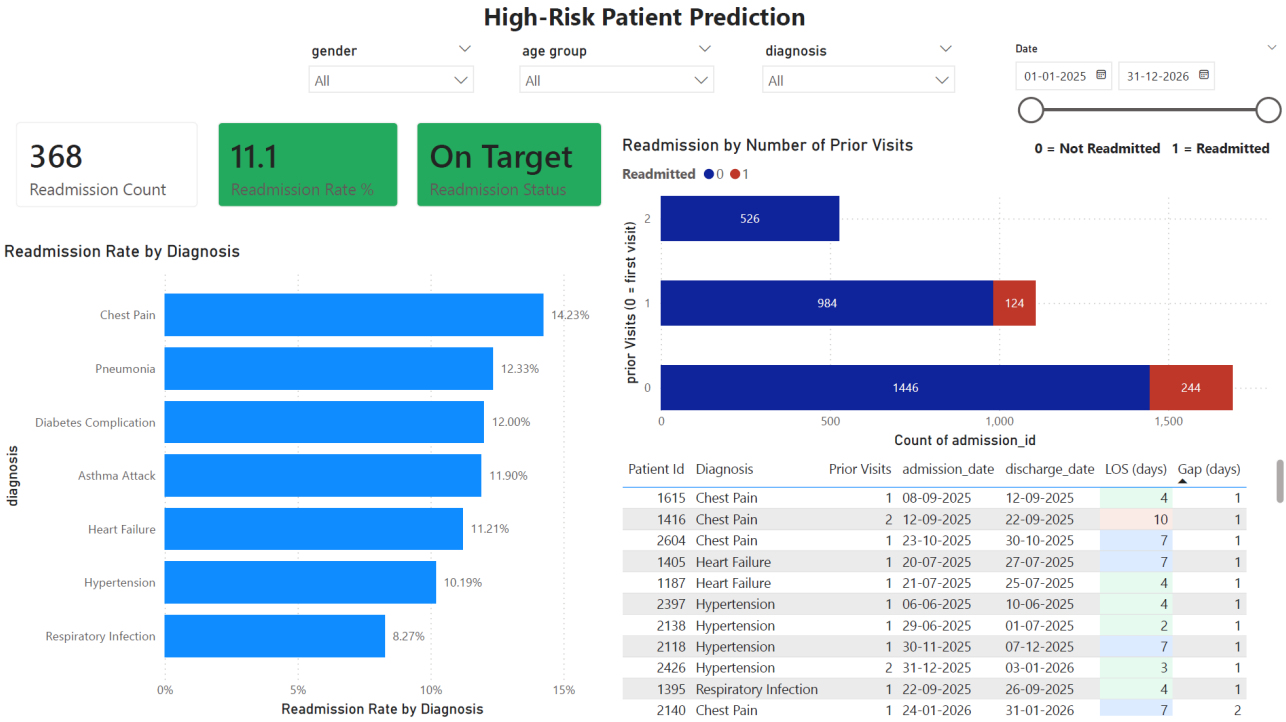
Average LOS of 6.0 days is 1 day above the 5-day clinical target. Crucially, all seven diagnoses sit within a 0.3-day band (5.8–6.1 days), which tells us this is a hospital-wide discharge process issue — not a condition-specific outlier that can be fixed by changing protocol for one diagnosis.

Discharge volumes track admission volumes closely in most months, indicating the hospital is not building up a backlog. March 2026 shows a higher discharge count than admissions because patients admitted in late February are being discharged in March.

Page 2: High-Risk Patient Identification

“Who is most likely to come back?”

The 11.1% readmission rate card is green because it is below the 15% target; but the bar chart below shows Chest Pain and Pneumonia exceeding the hospital average, telling the clinical team where to focus their discharge planning. The Prior Visits stacked bar is the key analytical insight: the red (readmitted) segment grows proportionally as prior visits increase, directly validating the ML model's top feature importance finding.



Patient Id	Diagnosis	Prior Visits	admission_date	discharge_date	LOS (days)	Gap (days)
1615	Chest Pain	1	08-09-2025	12-09-2025	4	1
1416	Chest Pain	2	12-09-2025	22-09-2025	10	1
2604	Chest Pain	1	23-10-2025	30-10-2025	7	1
1405	Heart Failure	1	20-07-2025	27-07-2025	7	1
1187	Heart Failure	1	21-07-2025	25-07-2025	4	1
2397	Hypertension	1	06-06-2025	10-06-2025	4	1
2138	Hypertension	1	29-06-2025	01-07-2025	2	1
2118	Hypertension	1	30-11-2025	07-12-2025	7	1
2426	Hypertension	2	31-12-2025	03-01-2026	3	1
1395	Respiratory Infection	1	22-09-2025	26-09-2025	4	1
2140	Chest Pain	1	24-01-2026	31-01-2026	7	2

Key Findings: Page 2

368 of 3,324 admissions (11.1%) resulted in a 30-day readmission. This is below the 15% clinical target; but the headline number masks meaningful variation across diagnoses.

Chest Pain has the highest readmission rate at 14.2%, followed by Pneumonia at 12.3%. Both exceed the hospital average of 11.1% and warrant targeted discharge planning interventions.

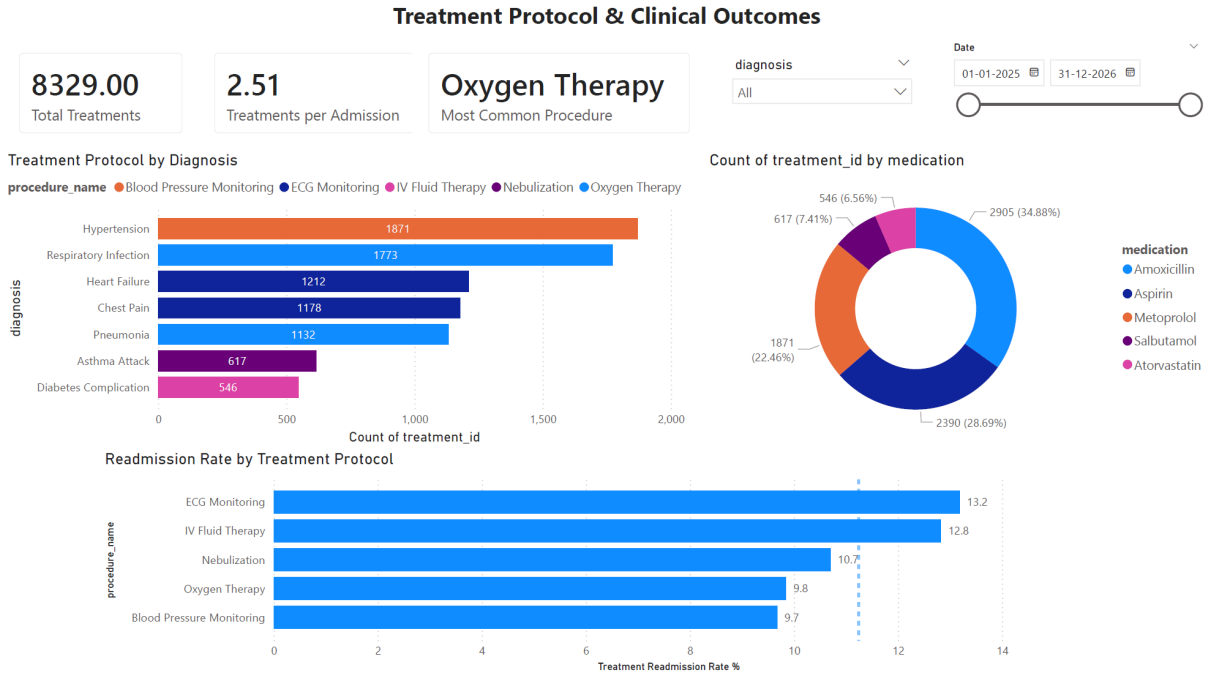
Respiratory Infection has the lowest rate at 8.3% despite being the second-highest volume diagnosis; suggesting its treatment pathway includes more effective patient education or follow-up.

The Prior Visits chart is the most clinically significant finding on this page: patients returning for a second or third admission have a proportionally higher readmission rate than first-time patients. This directly validates prior_admissions as the top feature in the XGBoost model (28% importance weight). The implication is clear; repeat visitors need more intensive discharge support, not the same standard protocol.

Page 3: Treatment Protocol & Clinical Outcomes

“Is our treatment consistent and does it connect to outcomes?”

The protocol chart shows 100% standardisation, which is a governance finding. The readmission rate by treatment protocol chart shows ECG Monitoring (cardiac procedures) with the highest readmission rate at 13.2%; but this is explained in the insight card as a reflection of cardiac diagnosis severity, not treatment failure. This distinction is important and shows the difference between correlation and causation.



Key Findings: Page 3

All 8,329 treatment records show a perfect one-to-one mapping between diagnosis and protocol: every Hypertension patient receives Blood Pressure Monitoring + Metoprolol, every Asthma patient receives Nebulization + Salbutamol, with zero exceptions. This 100% protocol adherence is a clinical governance finding; it confirms that treatment decisions are standardised and rule-based.

ECG Monitoring (used for Chest Pain and Heart Failure) shows the highest associated readmission rate at 13.2%. However, this is not a treatment failure. It is a reflection of the underlying severity of cardiac diagnoses which carry higher readmission risk by nature. The treatment is doing its job; the risk comes from the condition, not the care.

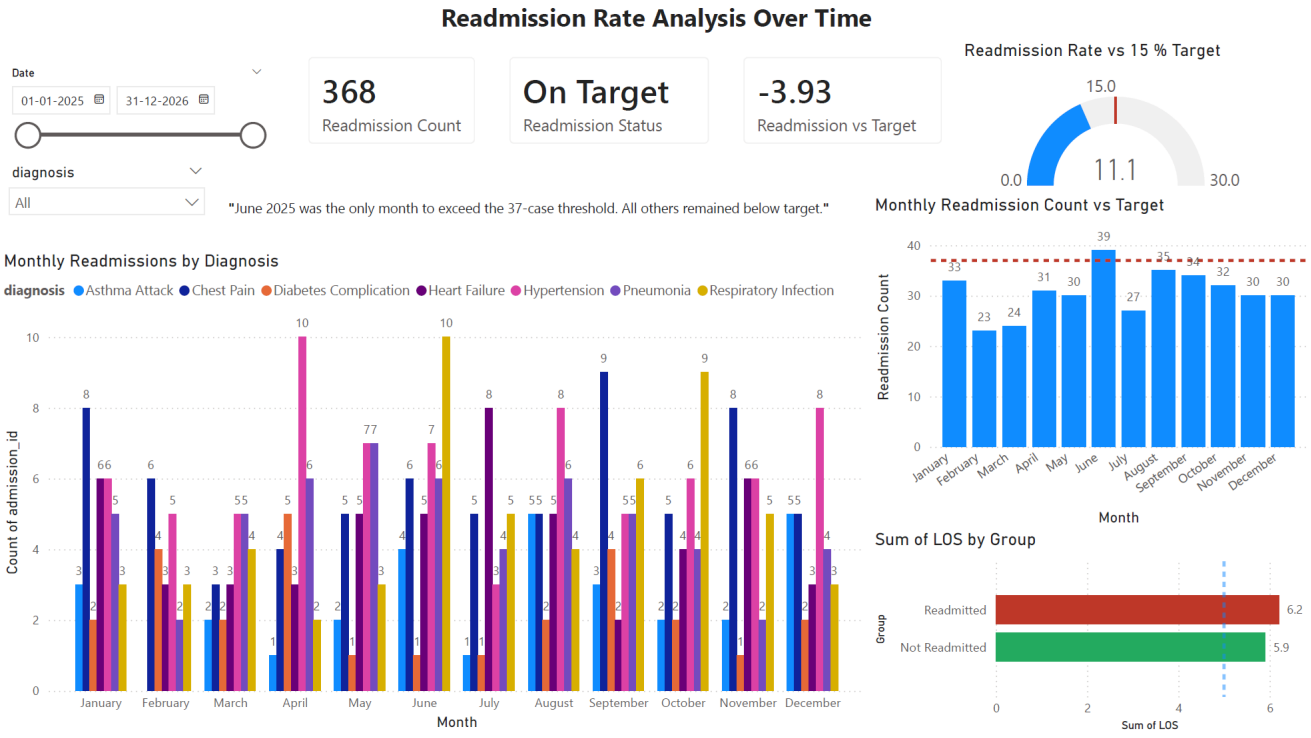
Oxygen Therapy, the most-used procedure (2,905 records across Pneumonia and Respiratory Infection), shows a mid-range readmission rate of 9.8%, consistent with those diagnoses' lower-than-average risk profiles.

The medication donut confirms the same pattern: Amoxicillin (antibiotic for respiratory infections) accounts for 34.9% of all medication events, followed by Aspirin for cardiac cases at 28.7%.

Page 4: Readmission Rate Analysis Over Time

“Are we improving and will we hold below the target?”

The gauge showing 11.1 vs 15% is the headline. The monthly count chart with the threshold line at 37 cases shows that only June 2025 breached the limit in 13 months. The LOS comparison chart connects this page back to Page 1: readmitted patients average 6.2 days vs 5.9 days for non-readmitted patients, giving the operations team a causal thread to investigate.



Key Findings: Page 4

The 30-day readmission rate of 11.1% sits 3.93 percentage points below the 15% clinical target; a meaningful margin that confirms the hospital's discharge and follow-up processes are working.

11 of 12 months remained below the 37-case threshold (equivalent to 15% of average monthly admissions). June 2025 was the only breach at 39 cases; an isolated spike with no sustained trend following it. This rules out a systemic deterioration and points to a transient cause worth investigating in the June data specifically.

Readmitted patients average 6.2 days LOS compared to 5.9 days for non-readmitted patients. This 0.3-day gap connects the LOS finding from Page 1 to the readmission outcome; longer stays are a mild predictor of readmission risk, suggesting discharge planning for patients approaching day 6 or beyond should receive additional scrutiny.

The monthly breakdown by diagnosis shows Hypertension consistently generating the most readmission events in absolute terms (due to its high admission volume), while Heart Failure generates fewer total events but at a higher rate relative to its volume.

Page 5: Post-Discharge Follow-Up Tracker

“Who do we call today, in what order?”

It is the only prescriptive page in the dashboard; it does not describe the problem, it solves it. The Priority by Diagnosis stacked bar shows Heart Failure generating the highest proportion of URGENT cases. The worklist table shows 358 readmitted patients sorted by gap days, with a traffic-light colour system and an Action Required column that tells each coordinator exactly what to do. Patients with a 1-day gap are at the top, in red, requiring a call within 24 hours.

Post-Discharge Follow-Up Tracker

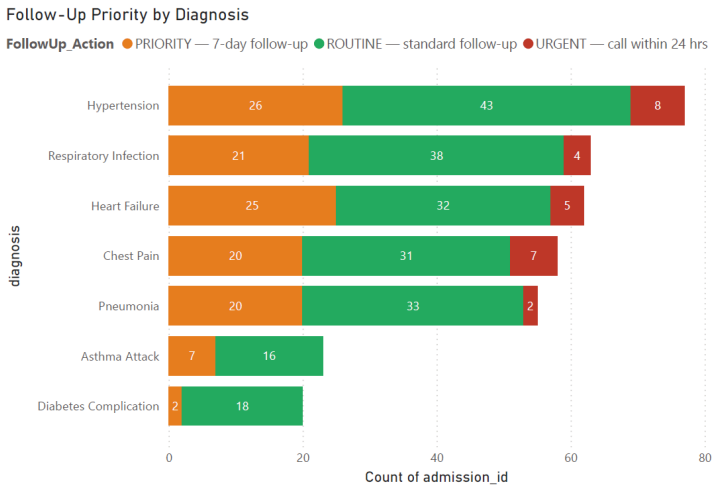
74 Discharged Last 7 Days

26 Urgent Follow Up

358 Needs Follow Up

diagnosis: All

Date: 01-01-2025 – 31-12-2026



Patient Id	Diagnosis	Discharge Date	Gap (days)	Action Required
1615	Chest Pain	12-09-2025	1	URGENT — call within 24 hrs
1416	Chest Pain	22-09-2025	1	URGENT — call within 24 hrs
2604	Chest Pain	30-10-2025	1	URGENT — call within 24 hrs
1187	Heart Failure	25-07-2025	1	URGENT — call within 24 hrs
1405	Heart Failure	27-07-2025	1	URGENT — call within 24 hrs
2397	Hypertension	10-06-2025	1	URGENT — call within 24 hrs
2138	Hypertension	01-07-2025	1	URGENT — call within 24 hrs
2118	Hypertension	07-12-2025	1	URGENT — call within 24 hrs
2426	Hypertension	03-01-2026	1	URGENT — call within 24 hrs
1395	Respiratory Infection	26-09-2025	1	URGENT — call within 24 hrs
2140	Chest Pain	31-01-2026	2	URGENT — call within 24 hrs
2197	Chest Pain	12-03-2026	2	URGENT — call within 24 hrs
1338	Heart Failure	20-06-2025	2	URGENT — call within 24 hrs
2327	Heart Failure	06-11-2025	2	URGENT — call within 24 hrs
1663	Heart Failure	07-02-2026	2	URGENT — call within 24 hrs
1521	Respiratory Infection	31-10-2025	2	URGENT — call within 24 hrs
1373	Chest Pain	03-06-2025	3	URGENT — call within 24 hrs
2609	Chest Pain	19-11-2025	3	URGENT — call within 24 hrs
1729	Hypertension	18-04-2025	3	URGENT — call within 24 hrs
2521	Hypertension	01-09-2025	3	URGENT — call within 24 hrs
2345	Hypertension	24-11-2025	3	URGENT — call within 24 hrs

Key Findings: Page 5

358 patients are in the active follow-up pipeline. Of these, 26 were readmitted within 3 days of discharge; these are the URGENT cases requiring same-day coordinator outreach. A gap of 3 days or fewer between discharge and readmission indicates the patient was not ready for discharge or lacked adequate post-discharge support.

Heart Failure generates the highest proportion of URGENT cases relative to its total volume. This is consistent with its 13.1% readmission rate on Page 2 and its highest average ML risk score on the model feature importance chart.

Diabetes Complication generates the fewest follow-up cases both in total volume and urgency; its 9.8% readmission rate and standardised IV Fluid + Atorvastatin protocol appear to be working effectively.

The worklist table is sorted by gap days ascending, ensuring coordinators always see the most urgent patients first. Filtering by diagnosis generates a ward-specific action list; for example, filtering to Heart Failure gives the cardiology team their own prioritised daily call sheet.

The three-tier SLA system (URGENT: 24hrs, PRIORITY: 7 days, ROUTINE: standard) replaces the manual spreadsheet process that previously had no consistent prioritisation logic.

6. Results & Business Impact

The solution achieved its primary goal. The 30-day readmission rate of 11.1% is 3.9 percentage points below the 15% clinical target, and 11 of 12 months stayed below that threshold throughout the reporting period. June 2025 was the single breach, reaching 39 cases against a 37-case threshold; an isolated spike rather than a trend.

The post-discharge follow-up tracker directly replaced the manual coordinator process. Instead of maintaining a spreadsheet, coordinators now open Page 5 each morning and work through the URGENT rows first. The 26 patients with a gap of 3 days or fewer represent the highest-risk cohort and would previously have required manual identification from raw admission logs.

The one open item is Length of Stay. At 6.0 days against a 5-day target, and with readmitted patients averaging 6.2 days compared to 5.9 for non-readmitted patients, extended stays are a risk factor that the hospital can act on. Targeting discharge planning interventions at patients approaching day 6 is the recommended next step.